

Project Workflow

The **Smart Lender – Loan Approval Prediction** project follows a structured workflow consisting of five major epics. Each epic represents a key stage in the machine learning lifecycle, beginning with data collection and ending with the deployment of a web-based application. This systematic approach ensures efficient project development and accurate loan prediction.

Epic 1: Data Collection and Architecture Design

Story 1: Dataset Collection

- Download the loan prediction dataset from the provided source.
- Store the dataset in the project directory for further processing.

Story 2: Architecture Design

- Design the overall system architecture.
 - Define the workflow for data processing, machine learning model development, and application deployment.
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Epic 2: Data Visualization and Analysis

Story 1: Data Import

- Import the dataset using the Pandas library.
- Examine the dataset structure and data types.

Story 2: Univariate Analysis

- Analyze the distribution of individual features using statistical and graphical methods.

Story 3: Bivariate Analysis

- Study the relationships between two variables to identify important correlations.

Story 4: Multivariate Analysis

- Explore relationships among multiple variables to discover hidden patterns and insights.
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Epic 3: Data Pre-processing

Story 1: Data Cleaning

- Identify and handle missing values, duplicate records, and inconsistencies in the dataset.

Story 2: Data Balancing

- Balance the dataset to ensure equal representation of target classes and improve model performance.

Story 3: Feature Scaling

- Normalize and scale numerical features to improve the efficiency and accuracy of machine learning algorithms.

Story 4: Data Splitting

- Split the processed dataset into training and testing sets for model development and evaluation.
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Epic 4: Model Building

Story 1: Decision Tree Model

- Train and evaluate a Decision Tree Classifier using the processed dataset.

Story 2: Random Forest Model

- Build and test a Random Forest Classifier to improve predictive performance.

Story 3: K-Nearest Neighbors (KNN)

- Train a KNN model and evaluate its prediction accuracy.

Story 4: XGBoost Model

- Implement an XGBoost classifier and compare its performance with the other machine learning models.

Story 5: Model Evaluation

- Compare all models using performance metrics such as Accuracy, Precision, Recall, and F1-Score.
 - Save the best-performing model for deployment in the web application.
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Epic 5: Application Building

Story 1: User Interface Development

- Design and develop responsive HTML pages to provide a user-friendly interface.

Story 2: Flask Application Development

- Build the Flask application and integrate the trained machine learning model for prediction.

Story 3: Testing and Deployment

- Run, test, and validate the application to ensure accurate loan predictions and smooth functionality before deployment.